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LID CONSTRUCTION FOR BREATHING APPARATUS

Abstract

A breathing assistance apparatus has a lid and latch assembly configured for one- handed operation. The apparatus also includes a hinge assembly that separates when over- rotated but that reengages upon closing of the lid onto the main body of the apparatus.

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Background/Summary

[0001] INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

[0002] Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under **37 CFR 1.57**.

BACKGROUND OF THE INVENTION

Field of the Invention

[0003] The present invention generally relates to respiratory devices. More particularly, the present invention relates to respiratory devices in which respiratory gases are supplied at a positive pressure after passing over a surface of liquid in a reservoir.

Description of the Related Art

[0004] CPAP treatment of obstructive sleep apnea involves the delivery of pressurized, breathable gas, usually air, to a user's airways using a conduit and a user interface, such as a mask. The gas pressures employed for CPAP typically range from about 4 cm H₂O to about 28 cm H₂O at flow rates of up to about 180 L/min (measured at the user interface), depend upon the requirements of the user. The pressurized gas acts as a pneumatic splint for the airway of the user. As such, the pressurized gas reduces the likelihood of collapsing of the airway.

SUMMARY OF THE INVENTION

[0005] CPAP machines include an airflow generator to supply the pressurized gas and many CPAP machines include a heated water bath or other source of water for humidification of the pressurized gas. CPAP machines often are used in bedrooms or other sleeping quarters and are placed on nightstands, for example. As such, reducing the footprint of CPAP machines is desirable. In addition, given the limited space of nightstand tops, CPAP machines often are pushed as close to a wall as possible.

[0006] Because of the cramped spaces in which CPAP machines are used, easy manipulation of a lid or other component used to enclose a water reservoir or the like is desired. Preferably, the lid or other such component can be easily pivoted about hinges. More preferably, the hinges are configured to be generally flush with, or recessed into, the adjoining outer surfaces. Even more preferably, the hinges provide a restraint against forces generated by air pressure within the machine (e.g., vertical forces with a generally horizontal lid) while allowing detachment of the lid when the lid is overextended during opening.

[0007] Moreover, to improve the ability to access the water reservoir, the lid preferably carries a

latch mechanism such that a user can open the lid with a single hand. In other words, the latch can be operated and the lid can be opened with a single hand and, preferably, with the single hand in a single position. Such configurations are a welcomed improvement over configurations requiring one hand to operate the latch and another hand to subsequently raise the lid.

[0008] In some configurations, a breathing assistance apparatus comprises a lid and a main body with the lid and the main body selectively enclosing a cavity. The lid comprises a button member and at least one hinge assembly.

[0009] In some configurations, the lid further comprises a component adapted to be engaged with one or more fingertips while the button member is depressed.

[0010] In some configurations, the button member is spring biased away from the at least one hinge assembly.

[0011] In some configurations, the at least one hinge assembly allows pivotal movement of the lid relative to the main body.

[0012] In some configurations, the hinge assembly comprises at least one post and at least one support, the supports overlying the post and the post being positioned inboard of an outer perimeter of the lid.

[0013] Preferably, the hinge assembly comprises a clip extending along at least an axial length of the post, the clip being positioned on an opposite side of the post from the supports such that the post is captured between the clip and the supports.

[0014] In some configurations, the at least one hinge assembly comprises a guide structure and cam structure, the guide structure and the cam structure configured to cause sliding movement of the post during rotation of the post.

[0015] In some configurations, the guide structure comprises a guide surface, wherein in use the guide surface contacts the cam structure to move the lid from a detached state to an attached state relative to the main body via the sliding movement of the post as a result of closing motion of the lid.

[0016] Preferably, the sliding movement only occurs when the lid is moved from a disengage state to an engaged state with respect to the main body.

[0017] Preferably, the sliding movement of the post is substantially horizontal with the supports being positioned at least vertically above the post with the lid in the closed position.

[0018] In some configurations, the at least one support defines an opening that permits the at least one post to pass through the opening such that in use the lid can be moved from a detached state to an attached state relative to the main body.

[0019] In some configurations, a method of opening a lid of a breathing apparatus comprises moving a button member toward an engagement component to unlatch the lid and then rotating the lid to an opened position without the user having to reposition a hand.

[0020] In some configurations, a method of engaging a lid on a main body of a breathing apparatus comprises placing a post alongside at least one support member with the lid in an open position, pivoting the lid toward a closed position such that at least one engaging structures formed on the lid and at least one engaging structure formed on the main body cause the lid to translate into an engaged structure with the lid being pivotable while in the engaged structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] These and other features, aspects and advantages of the present invention will be described with reference to the following drawings.

[0022] FIG. 1 is a rear perspective view of a breathing assistance apparatus that is arranged and configured in accordance with certain features, aspects and advantages of the present invention.

[0023] FIG. 2 is an enlarged rear top perspective view of a hinge of the breathing assistance apparatus of FIG. 1 with a lid in a closed position.

[0024] FIG. 3 is an enlarged front perspective view of the hinge of FIG. 2 showing the lid in an opened position.

[0025] FIG. 4 is a sectioned perspective view of the hinge of FIG. 2 showing the lid in a partially closed position.

[0026] FIG. 5 is a front perspective view of the lid of the breathing assistance apparatus of FIG. 1, which lid is arranged and configured in accordance with certain features, aspects and advantages of the present invention.

[0027] FIG. 6 is a rear bottom perspective view of a latch member of the lid of FIG. 5.

[0028] FIG. 7 is a rear bottom perspective view of the latch member of FIG. 6 shown assembled to the lid.

[0029] FIGS. 8A-8D are side views of a guide structure of the hinge of FIG. 2 in several positions from opened to closed.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0030] FIG. 1 illustrates a breathing assistance apparatus **100** that is arranged and configured in accordance with certain features, aspects and advantages of the present invention. The apparatus **100** comprises a main body **102** and a lid **104**. Together, the main body **102** and the lid **104** define a housing **106** that encloses, or generally encases, a reservoir, tub, tank or other body of liquid (not shown) within a cavity.

[0031] The illustrated body **102** comprises at least one outer surface **110**. In the illustrated configuration, the body **102** comprises four generally planar outer surfaces **110** that are connected by rounded corners **112**. Other configurations are possible.

[0032] The lid **104** is connected to the body **102** with hinge assemblies **114**. In the illustrated configuration, the lid **104** is connected to a rear of the main body **102** using two hinge assemblies. The lid **104** can be connected to other surfaces. In addition, the lid **104** can be connected to the main body **102** using as few as one hinge assembly **114** or more than two hinge assemblies **114**. Preferably, the hinge assemblies **114** are constructed such that, with the lid **104** in the closed position (e.g., as shown in FIG. 1), the hinge assemblies **114** are generally flush with or recessed into the rear outer surface **110**. In some configurations, the hinge assemblies **114** are constructed such that the hinge assemblies do not protrude rearward of the rear outer surface **110**. In some configurations, some of the hinge features may protrude very slightly from one or more surrounding surface of the rear outer surface or other surrounding portion of the apparatus. In some configurations, the lid **104** comprises an outer perimeter **116** and the hinge assemblies **114** do not protrude significantly outward of the outer perimeter **116** of the lid **104**.

[0033] With reference to FIG. 2, the hinge assemblies **114** comprise at least one post **120** and at least one support **122**. Preferably, the at least one post **120** is mounted such that the post is recessed and located inside of the outer perimeter **116** of the lid. The illustrated configuration comprises a single post **120** that is connected to two supports **122**. In some configurations, the at least one post **120** is mounted to the lid **104** while the at least one support **122** is mounted to the main body **102**. In some configurations, the at least one post **120** can be mounted to the main body **102** while the at least one support **122** can be mounted to the lid **104**.

[0034] The at least one post **120** can be secured to the lid **104** using one or more standoffs **124**. In the illustrated construction, the standoffs **124** are positioned between the two supports **122**. The standoffs reduce the deflection of the post **120** in the region of the supports **122**. In some configurations, the portion **126** of the post **120** extending between the standoffs **124** can be omitted. In the illustrated configuration, however, the portion **126** of the post **120** can be secured by a clip **130**. The clip **130** can have a recess that accommodates the post **120** or any other suitable configuration.

[0035] A force generated by the clip **130** on the post **120** in the illustrated configuration is opposed

by forces generated by the supports **122**, for example. The clip **130** can contact an opposite side of the post **120** relative a contact region between the supports **122** and the post **120**. In some configurations, the clip **130** can be configured to deflect away from the post **120** during assembly of the post **120** and the clip **130**. For example, the clip **130** can be a deflectable finger that extends rearward from the main body **102**. The supports **122** can have a recess that defines a rearwardly-opening mouth that receives at least a portion of a circumference of the post **120**. Accordingly, movement of the post **120** in the directions shown by the arrow can result in attachment and detachment of the post **120** relative to the supports **122** and the clip **130**. The rearwardly-opening mouth of the supports **122** also results in a structure that overlies the post **120** such that vertical movement of the post **120** relative to the supports **122** is not possible with the post **120** fully received within the mouths of the supports **122**. As such, the lid **104** is unlikely to separate from the main body **102** during therapy.

[0036] With reference to FIG. 3, the lid **104** and the main body **102** can include structures that guide the post **120** and the clip **130** and supports **122** into connection. In some configurations, the structures facilitate movement of the at least one post **120** into the mouth of the supports **122** and into engagement with the clip **130** when the lid **104** is brought toward the closed position on the main body **102**. In the illustrated configuration, two guide structures **140** are positioned outboard of the two supports **122**. Other configurations also are possible.

[0037] With reference to FIG. 4, the guide structure **140** comprises a recess **142** that receives a cam structure **144**. The cam structure **144** is received within the recess **142** such that, with the lid **104** moved to the fully closed position on the main body **102**, the cam structure **144** does not prevent the lid **104** from fully closing. The cam structure **144** tapers toward a point with a sloping surface **146** such that the thickness of the cam structure **144** increases from back to front while the guide structure has an opposing surface **150**. The sloping surface **146** and the opposing surface **150** engage each other such that the sloping surface **146** and the opposing surface **150** resists movement of the lid **104** relative to the main body **102** in the detachment direction (see FIG. 2) when the lid **104** is in an at least partially closed position.

[0038] Moreover, as the lid **104** is rotated toward a closed position (e.g., FIG. 1), the sloping surface **146** and the opposing surface **150** act to draw the lid **104** forward into the attachment position (e.g., the at least one post **120** connected to the supports **122** and the clip **130**). Thus, rotation of the lid **104** can result in the hinge assemblies **114** being operatively connected from a detached state. In some configurations, the difference in thicknesses of the sloping surface **146** and the opposing surface **150** from initial contact to fully closed is substantially the same as a distance from the outer surface of the post **120** and the first contact with the supports **122** such that as the sloping and opposing surfaces **146**, **150** are moved into full engagement along their lengths, the post **120** is pulled into position within the supports **122** and the clip **130**.

[0039] Notably, with the lid **104** in the open position (e.g., FIG. 3) relative to the main body **102**, sufficient space exists between the sloping surface **146** and the opposing surface **150** such that the lid **104** can be separated from the main body **102** with the application of force. Advantageously, such a construction allows separation between the lid **104** and the main body **102** by disconnection of the posts **120** from the supports **122** and the clips **130**. Such a separation can protect the apparatus **100** from damage resulting from forces that result from over-opening of the lid or from possible misuse of the apparatus **100**. Accordingly, the ability to separate the lid **104** from the main body **102** can protect the apparatus **100** from permanent damage. As discussed above, the lid **104** can be easily reconnected by placing the posts **120** alongside the supports **122** and rotating the lid **104** toward a closed position, which results in the sloping surface **146** of the cam structure **144** and the opposing surface **150** of the guide structure drawing the components back together.

[0040] With reference now to FIG. 5, a release and handle assembly **160** of the lid **104** will be described. As illustrated, the release and handle assembly **160** comprises at least one button member **162** that is mounted on the lid **104**. In some configurations, the button member **162** is

mounted for movement toward and away from the hinge assemblies **162**. In some configurations, the button member **162** can be mounted along a side of the lid **104** that is adjacent to the side of the lid **104** having the hinge assemblies **162**. In the illustrated configuration, the button member **162** is mounted on an opposing side of the lid **104** relative to the hinge assemblies **114** and capable of movement toward and away from the hinge assemblies **114**.

[0041] The button member **162** is illustrated in FIG. **6**. The button member comprises an exterior surface **164**, which is best shown in FIG. **5**. The exterior surface **164** is sufficiently large for contact with a finger or thumb. As also illustrated in FIG. **5**, the lid **104** can comprise a component **166** that is used to raise the lid **104**. In the illustrated configuration, the component **166** is a recessed region that has a wall **170** extending generally parallel with the exterior surface **164**. Preferably, the recessed region **166** is sufficiently large as to accommodate at least tips of fingers. In some configurations, the component **166** is one or more lip, ridge, protrusion, recess or the like.

Accordingly, the component **166** (e.g., having the wall **170**) and the exterior surface **164** can allow a simple squeezing action to accomplish both an unlatching of the button member **162** as well as providing a grasping action to allow movement of the lid **104** away from the main body **102**.

[0042] With reference again to FIG. **6**, the illustrated button member **162** comprises two outer posts **172**. The posts **172** are sized and configured to engage with biasing members, such as compression springs or the like, for example but without limitation. The posts **172** are received within corresponding housings **174** that are formed in the lid **104**. The housings **174** can contain the biasing members and the posts **172** can slide into and out of the housings **174** against the biasing force of the biasing members.

[0043] The illustrated button member **162** also comprises at least one tab **176**. In the illustrated configuration, the button member **162** comprises two tabs **176**. The tabs **176** are bayonet shaped with a barb **180** at the end. As shown in FIG. **7**, the tabs **176** can engage a structure **182** formed in the lid **104**. Thus, the tabs **176** in cooperation with the structure reduce the likelihood of the button member **162** coming out of a recess of the lid **104** while allowing inward depression of the button member **162** relative to the lid **104**.

[0044] The structure **182** also comprises a passage **184** that receives a finger **186** mounted to, or formed on, the button member **162**. The passage **184** facilitates generally only linear translation of the button member **162** relative to the lid **104**. In other words, the passage **184** receives the finger **186** and the finger **186** is generally limited by the passage **184** to substantially linear movement.

[0045] The finger **186** supports a locking post **190**. The locking post **190** can engage with a structure formed on the main body **102**. In some configurations, the locking post **190** engages with a structure formed on an inner surface of the main body **102** such that the depression of the button member **162** results in separation between the locking post **190** and the structure with which it normally engages. Releasing of the button member **162** results in the biasing members moving the locking post **190** into position for reengagement with the structure with which it normally engages.

[0046] FIGS. **8A-8D** illustrate the lid **104** in several positions relative to the main body **102**. FIG. **8A** illustrates the lid **104** in a first opened position. FIGS. **8B** and **8C** illustrate the lid **104** in progressively further closed positions relative to the first opened position of FIG. **8A**. FIG. **8D** illustrates the lid **104** in a closed position. In particular, FIGS. **8A-8D** further illustrate the interaction between the sloping surface, or cam surface, **146** of the cam structure **144** and the opposing surface (also referred to as a guide surface or cam follower surface) **150** of the lid recess **142** as the lid **104** moves from an opened position to a closed position. As described herein, the cam structure **144** and the recess **142** can interact to move the lid **104** from a detached position relative to the main body **102** to an attached position relative to the main body **102** when the lid **104** is moved from an opened position to a closed position.

[0047] As described herein, the support **122** includes the recess **200** comprising the mouth **202**, which opens in a rearward direction in the illustrated arrangement. Preferably, the recess **200** is elongate in shape and extends in a generally fore-aft direction. In the illustrated arrangement, the

recess **200** is generally or substantially horizontal or aligned with a bottom support surface of the main body **102** such that the recess **200** is generally or substantially horizontal when the apparatus **100** is rested on a flat surface. The recess **200** includes upper and lower guide surfaces **204** that extend forwardly from the mouth **202** to an end surface **206**. As described herein, the support **122** also assists in retaining the lid **104** on the main body **102** in response to forces within the housing **106** (e.g., forces resulting from internal pressure) tending to separate the lid **104** from the main body **102**. Thus, the upper portion of the end surface **206** and/or the upper guide surface **204** can form a retention surface that contacts the post **120** to retain the lid **104**, or at least a portion of the lid **104** near the support **122** and/or hinge **114** (e.g., a rearward portion), on the main body **102** in response to forces tending to separate the lid **104** from the main body **102**.

[0048] In the illustrated arrangement, the end surface **206** is curved (e.g., semi-circular) in shape from a side view and connects the upper guide surface **204** and the lower guide surface **204**. The illustrated support **122** is elongate in a lateral or side-to-side direction and, thus, defines a substantially semi-cylindrical shape in three dimensions. However, for convenience, the hinge **114**, guide structure **140** and cam structure **144** may be described herein in the context of the side views of FIGS. **8A-8D** in two-dimensional terms. It will be appreciated that the described structures also have a width dimension relative to the apparatus **100** or a depth dimension relative to the side view of FIGS. **8A-8D**.

[0049] The illustrated end surface **206** is defined by a curve having a radius **208** from a center point **210** (or an axis in three dimensions). The recess **200** can have a longitudinal axis **212** that extends along a length of the recess **200** generally from the mouth **202** to the end surface **206** and, in the illustrated configuration, passes through the center point **210**. The axis **212** can be aligned with or parallel to one or both of the upper and lower guide surfaces **204** or can be centrally-located between the guide surfaces **204** (such as in the event of a tapered recess, for example and without limitation). In the illustrated configuration, the axis **212** is generally or substantially horizontal (or parallel to a bottom surface of the main body **102**). However, in other configurations, the axis **212** could be non-horizontal or angled relative to the bottom surface of the main body **102**).

[0050] As described herein, the post **120** preferably is cylindrical in shape or circular in shape from a side view, as illustrated in FIGS. **8A-8D**. The post **120** defines a center point **214** (or an axis in three dimensions) and a radius **216**. Preferably, the radius **216** of the post **120** is substantially equal to the radius **208** of the end surface **206** such that the post **120** can be snugly positioned against the end surface **206** and restrained in a vertical direction within recess **200** with the center points **210**, **214** substantially aligned or coaxial with one another.

[0051] Preferably, the cam surface or sloping surface **146** of the cam structure **144** is positioned or oriented relative to the end surface **206** and/or center point **210** such that the post **120** is moved along the axis **212** of the recess **200** toward the end surface **206** as the lid **104** is moved from the opened toward the closed position. In the illustrated arrangement, a first distance **220** is defined between a first point or location **222** on the sloping surface **146** and the center point **210** and a second distance **224** is defined between a second point or location **226** on the sloping surface **146** and the center point **210**. Preferably, the first point **222** is located on an initial portion of the sloping surface **146** and the second point **226** is located on a subsequent portion of the sloping surface **146** such that the opposing surface **150** will contact the first point **222** before the second point **226** as the lid **104** is moved from an opened position to a closed position. The first point **222** can be at or near the location that is first contacted by the opposing surface **150** during closing of the lid **104** and the second point **226** can be at or near the furthest location along the sloping surface **146** contacted by the opposing surface **150** when the lid **104** is in the closed position. The first distance **220** preferably is smaller than the second distance **224**, which causes the movement of the post **120** along the axis **212** of the recess **200** as the lid **104** is moved from an opened position to a closed position and the opposing surface **150** moves along the sloping surface **146** in a direction from the first point **222** towards the second point **226**.

[0052] A distance **228** traveled by the center point **214** of the post **120** within the recess **200** when the lid **104** is closed preferably is substantially equal to a difference **230** between the second distance **224** and the first distance **220**. The distance **228** can be less than an overall length of the recess **200** because the post **120** can be at least partially located within the recess **200** prior to being drawn to the end surface **206** by the guide structure **140**. In other arrangements, the distance **228** could be substantially equal to a length of the recess **200**. Although the distances described immediately above are in relation to the center points **210** and **214**, the distances similarly could be measured relative to other points on locations on the recess **200** (e.g., end surface **206**) and post **120**, respectively, if desired, such as in the context of non-circular (or non-cylindrical) recess or post shapes.

[0053] FIG. **8A** illustrates the lid **104** in a first open position relative to the main body **102**, which may be a fully opened position, such as when initiating a process of reattachment of the lid **104**. As illustrated, the opposing surface **150** is clear of the sloping surface **146** such that rearward movement (e.g., detachment) of the lid **104** is permitted. Advantageously, such an arrangement reduces the likelihood of permanent damage resulting from excessive opening forces applied to the lid **104**. FIG. **8B** illustrates the lid **104** in a second open position that is further towards a closed position relative to the first open position of FIG. **8A**. In FIG. **8B**, the opposing surface **150** remains clear of the sloping surface **146**. That is, a vertical gap exists between a bottom of the opposing surface **150** and a top of the sloping surface **146**.

[0054] FIG. **8C** illustrates the lid **104** in a third open position that is further towards a closed position relative to the positions of FIGS. **8A** and **8B**. In the position of FIG. **8C**, a leading portion or engagement portion **232** of the opposing surface **150** is engaged with or contacts an initial portion of the sloping surface **146**, which may be at or near the first point or location **222**. In the illustrated arrangement, the post **120** is spaced away from the end surface **206** of the recess **200**. FIG. **8D** illustrates the lid **104** in a position that is further towards a closed position than the positions of FIGS. **8A-8C**. The position of FIG. **8D** can be a fully closed position of the lid **104**. In the illustrated arrangement, at least the leading portion or engagement portion **232** of the opposing surface **150** is engaged with or contacts a subsequent portion of the sloping surface **146** spaced from the initial portion. The subsequent portion may be at or near the second point or location **226**. Preferably, the post **120** is snugly positioned against the end surface **206** in the position of FIG. **8D**. Thus, preferably, a distance between the engagement surface **232** and the center point **214** is preferably substantially equal to the second distance **224**.

[0055] As described herein, the interaction of the guide structure **140** and the cam structure **144** as illustrated in FIGS. **8A-8D** may occur only during a reattachment or engagement procedure for the lid **104**. In normal opening and closing movement, sliding of the post **120** (or significant sliding of the post **120**) within the recess **200** may not occur. Advantageously, such an arrangement reduces wear by reducing the amount of sliding movement between components during normal use. However, other arrangements are possible in which the sliding movement of the post **120** occurs more often than only during engagement of the lid **104**, such as during normal opening and closing of the lid **104**.

[0056] Although the illustrated opposing surface **150** has a similar size and shape as the sloping surface **146** and extends along a substantial portion of the sloping surface **146** when the lid **104** is in the closed position, other arrangements are also possible. For example, the opposing surface **150** could be a smaller surface, such as a cam follower surface, defining an engagement portion **232** that contacts only a point or small length of the sloping surface **146** in any one position. The engagement portion **232** could be rotatable such that it rolls along the sloping surface **146**, if desired. Moreover, as disclosed herein, the illustrated arrangements could be reversed such that the cam surface or sloping surface **146** is carried by the lid **104** and the opposing surface **150** (or cam follower with engagement portion **232**) is carried by the main body **102**.

[0057] Unless the context clearly requires otherwise, throughout the description and the claims, the

words “comprise”, “comprising”, and the like, are to be construed in an inclusive sense as opposed to an exclusive or exhaustive sense, that is to say, in the sense of “including, but not limited to”.

[0058] Reference to any prior art in this specification is not, and should not be taken as, an acknowledgement or any form of suggestion that prior art forms part of the common general knowledge in the field of endeavour in any country in the world.

[0059] The invention may also be said broadly to consist in the parts, elements and features referred to or indicated in the specification of the application, individually or collectively, in any or all combinations of two or more of said parts, elements or features.

[0060] Where, in the foregoing description reference has been made to integers or components having known equivalents thereof, those integers are herein incorporated as if individually set forth.

[0061] It should be noted that various changes and modifications to the presently preferred embodiments described herein will be apparent to those skilled in the art. Such changes and modifications may be made without departing from the spirit and scope of the invention and without diminishing its attendant advantages. It is therefore intended that such changes and modifications be included within the scope of the invention.

Claims

1. (canceled)
2. A breathing assistance apparatus comprising: a main body comprising a cavity; a lid comprising: a top surface comprising an opening configured to receive at least one or more of a user's fingertips, a front portion comprising a recess for receiving a button member, wherein the button member has an exterior surface with a length to accommodate a user's thumb, a back portion positioned on an opposing side to the front portion of the lid, and wherein the lid is configured to rotate between an opened position and a closed position, wherein the closed position is configured to close the cavity; a hinge assembly positioned on the back portion of the lid, wherein the hinge assembly movably connects the lid to the main body; wherein the lid is secured to the main body when the button member is not engaged; wherein the button member is configured to move relative to the recess in a direction toward and away from the hinge assembly; and wherein the lid is movable between the closed position and an open position by a user simultaneously applying a force to engage the button member and securing the opening with at least one or more of the user's fingertips.
3. The breathing assistance apparatus of claim 2, wherein the button member is spring-biased away from the hinge assembly.
4. The breathing assistance apparatus of claim 2, wherein the button member comprises an internal structure configured to engage a housing positioned on an underside of the lid, wherein the internal structure is movable relative to the housing, and wherein the housing comprises a compression member for engaging with the internal structure.
5. The breathing assistance apparatus of claim 4, wherein the compression member comprises a spring.
6. The breathing assistance apparatus of claim 4, wherein the internal structure comprises at least one post extending within the housing and configured to move relative to the housing.
7. The breathing assistance apparatus of claim 4, wherein the internal structure comprises a retention structure configured to retain the button member within the recess of the front portion of the lid.
8. A breathing assistance apparatus comprising: a main body comprising a cavity; a lid comprising: a top surface comprising an engagement portion configured to receive at least one or more of a user's fingertips, a front portion comprising a recess for receiving a button member, a back portion positioned on an opposing side to the front portion of the lid, and wherein the lid is configured to

rotate between an opened position and a closed position, wherein the closed position is configured to close the cavity; a hinge assembly positioned on the back portion of the lid, wherein the hinge assembly movably connects the lid to the main body; and wherein the button member is configured to move relative to the recess in a direction toward and away from the hinge assembly.

9. The breathing assistance apparatus of claim 8, wherein the lid is secured to the main body when the button member is not engaged.

10. The breathing assistance apparatus of claim 8, wherein the lid is movable between the closed position and an open position by a user simultaneously applying a force to engage the button member and securing the engagement portion with at least one or more of the user's fingertips.

11. The breathing assistance apparatus of claim 8, wherein the button member has an exterior surface with a length to accommodate a user's thumb.

12. The breathing assistance apparatus of claim 8, wherein the engagement portion comprises a lip, a ridge, a protrusion, or a recess.

13. The breathing assistance apparatus of claim 8, wherein the recess provides a linear translation of the button member relative to the lid.

14. The breathing assistance apparatus of claim 8, wherein the button member is spring-biased away from the hinge assembly.

15. The breathing assistance apparatus of claim 8, wherein the button member comprises an internal structure configured to engage a housing positioned on an underside of the lid, wherein the internal structure is movable relative to the housing, and wherein the housing comprises a compression member for engaging with the internal structure.

16. The breathing assistance apparatus of claim 15, wherein the compression member comprises a spring.

17. The breathing assistance apparatus of claim 15, wherein the internal structure comprises at least one post extending within the housing and configured to move relative to the housing.

18. The breathing assistance apparatus of claim 15, wherein the internal structure comprises a retention structure configured to retain the button member within the recess of the front portion of the lid.

19. The breathing assistance apparatus of claim 8, wherein the recess is positioned closer to the front portion of the lid than the back portion of the lid.)

20. The breathing assistance apparatus of claim 8, wherein the recess is positioned to allow a hand of a user to engage both the recess and the button member simultaneously.)

21. The breathing assistance apparatus of claim 8, wherein the recess comprise walls that are flat.

22. The breathing assistance apparatus of claim 8, wherein the recess comprises a rearward facing wall.

23. A lid of a breathing assistance apparatus, the lid and a main body selectively enclosing a cavity configured to hold a body of liquid, the lid comprising: a top surface comprising an engagement portion; a front portion comprising a recess to receive a button member; a back portion positioned on the lid opposite of the front portion of the lid, wherein the back portion is configured to receive a hinge assembly for securing the lid to the breathing assistance apparatus; wherein the button member is positioned on an opposing side of the lid relative to the hinge assembly; and wherein the button member is configured to move in a direction toward and away from the hinge assembly.

24. The breathing assistance apparatus of claim 23, wherein the lid is secured to the main body of the breathing assistance apparatus when the button member is not engaged, and wherein the lid is disengaged from the main body of the breathing assistance apparatus by a user simultaneously applying a force to engage the button member and securing the engagement portion with at least one or more of a user's fingertips.

25. The breathing assistance apparatus of claim 23, wherein the button member comprises an internal structure comprising: at least one post extending within a housing positioned on an underside of the lid, wherein the at least one post is movable relative to the housing, a compression

member positioned within the housing and is configured to engage the at least one post, and a retention structure configured to retain the button member within the recess of the front portion of the lid.
